

omicstra TNBC-92

omicstra

project data: TNBC-92 (v3 gpath2vec retrain)

Cohort: Wang et al. 2024 TNBC, 94 patients, 38 held-out subarrays, 14 held-out test patients. v3 niche-join 208,786 niches (67,131 dropped: no significant pathway under MAD/mean gene selection). Test set: 35,594 niches. Unit: niche (k=6 spatial neighbors), ~1200 cells. Evaluation: subarray-level, patient-stratified 85/15 split, seed=42. FDR control: BH-FDR over the test families described per hypothesis. Numbers verified against `runs/tnbc-92_v3/eval/*` and `notebooks/final/05_summary_umaps_v3.ipynb` §9, §10.

v1 -> v3 single change. `gpath2vec_niche` (512-d) is the v3 build (`fisher_madmean_low_dim512_e5_s1234`, sha 13985cbd...) on the ST input. `novae`, `virchow2`, `mc_weights` unchanged. v3 adds R5 (AnInfoNCE + late fusion) as one additional contrastive method in the grid; it is evaluated identically to R1-R4/R6 and is not a separate experiment.

hypothesis verdict summary

proposal hypothesis	v3 verdict	how
H1 contrastive alignment improves cross-modal retrieval and representation alignment over unaligned baselines	supported	all 6 contrastive runs (R1-R6) beat the random-init B4 (chance, 0.496) and the unaligned-PCA B3 (sub-chance, 0.441) on AUC. cross-attention R4 (AUC 0.859, CKA 0.631)

proposal hypothesis	v3 verdict	how
H2 Part A archetype clusters more coherently in aligned space than in unimodal spaces	construct-validity correction applied, then supported	exceeds the proposal's 0.65-0.75 range; late-fusion runs land inside or slightly below it. the proposal-literal 9-archetype label is patient-level pseudobulk: $NMI(\text{archetype}, \text{patient_id}) = 0.89$; every niche in a patient inherits one archetype. archetype ARI is a patient-classification test in disguise. on the audit-correct per-spot 14-class <code>mc_megacluster</code> label, aligned space shows 5× lift over raw ST (0.046 -> 0.246). B.1 (categorical compartment cosine contrasts): TLS vs TIL-stroma passes in 5/9 runs (R1/R2/R5/R6/B2; v3 Welch $z = +2.2$ to $+5.9$). Tumor/ Necrosis contrasts invert (annotation granularity confound, not biology absence). B.2 (continuous TLS gene-signature CCA on z_{he}): 6/10 runs sig at BH-FDR<0.05 - B4 random-init strongest ($z=25.5$), B1
H2 Part B distinct compartments (tumor, TLS, necrosis) show higher matched-pair cosine than ambiguous stroma - same FTU tested two ways	partially supported, B.1 + B.2 disagree on the method ranking	

proposal hypothesis	v3 verdict	how
H3 Reactome pathway embeddings correlate with shared-latent directions via CCA, indicating preservation of interpretable biological signal	supported with cross-patient qualifier	CCA (16.6), R4 cross-attn (11.4), B2/B3 (11.3), R5 marginal (2.4); R1/R2/R3/R6 fail. Same FTU (TLS), two operationalizations, different method rankings - see Part B framing for what this means. within-cohort: pathway-direction CCA is positive across all runs (universal, non-discriminating). cross-patient sub-split (the discriminating operationalization): R4 4/4 testable pathways significant (z_A 11.2-25.4); classical baselines 4/4 with lower z_A; late-fusion contrastive runs 0-2/4. interpretable signal is present in R4 and classical baselines and transfers across held-out patients; late-fusion contrastive runs encode pathway signal that does not transfer.
systems-oriented does one alignment strategy dominate every biological	no, as predicted by the proposal framing	R4 wins H1 retrieval + H3 cross-patient transfer. R1/R5/R6 win H2 niche-level region clustering.

proposal hypothesis	v3 verdict	how
objective simultaneously?		classical baselines fail H1 retrieval but stay competitive on H3 hit-rate. the trade-off is the deliverable.

construct-validity note

The H2 Part A evaluation question as proposed - “*do the 9 spatial archetypes cluster more coherently in the aligned space*” - turned out not to be a good operationalization of the objective “*aligned latent captures TME organization.*” Reason: archetypes were assigned at patient level in the Wang dataset (every niche in patient P inherits archetype A_P), so “archetype ARI” mechanically resolves to a 14-class patient classifier at niche resolution. Classical baselines that amplify patient identity (B1 CCA: patient $z = 122.13$, $3.3\times$ amplification over raw H&E) score highest on the metric without encoding more biology. **The metric and the objective came apart - this is the construct-validity failure mode in concrete form.**

The correction is to evaluate H2 Part A on Wang’s per-spot 14-class `mc_megacluster` label (NMF on expression, niche-resolution, biology-derived) instead of the patient-level archetype label. On the corrected label the hypothesis is supported. The archetype-ARI numbers are reported in the parquet as diagnostic only and visualized in notebook §5 alongside `patient_id` panels for B1 and R4 - if a run’s archetype panel and its `patient_id` panel show identical structure, archetype ARI on that run is patient classification by construction. See `projects/tNBC-92/evaluation_question_audit.md` finding #1 for the formal audit.

The same logic guided two other decisions on H3 (use the cross-patient sub-split, not fit-on-all, because fit-on-all is universally supported and does not discriminate) and on `z_he` view selection (`v3 gpath2vec` is on the ST input, so `z_st` / `z_mean` are circular with the `gpath2vec`-derived signal; only `z_he` is non-circular and reported as headline).

method definitions

method	definition
R4	InfoNCE cross-attention alignment. ST queries attend over H&E tile tokens (center spot + 6 spatial neighbors at full Virchow2 resolution). produces a weighted-sum H&E representation conditioned on ST.
R1	InfoNCE late-fusion alignment with independent modality projections (mean-pooled H&E -> MLP, novae + gpath2vec -> MLP). standard CLIP-style contrastive baseline.
R5 (new in v3)	AnInfoNCE late-fusion alignment. extends InfoNCE with a per-dim learnable log-scale parameter on the bilinear inner product (diagonal Mahalanobis). evaluated identically to R1 across H1/H2/H3.
R6	R1 ablation: drops gpath2vec embeddings from the ST input. tests whether H&E pathway signal depends on the pathway anchor being present on the ST side.
R2	Supervised contrastive late-fusion alignment using soft mc_weights similarity as targets (instead of binary match/mismatch).
R3	Barlow Twins late-fusion: redundancy-reduction loss that pushes the cross-correlation matrix toward identity. no explicit negatives.
B1	Canonical Correlation Analysis. classical, linear, closed-form.

method	definition
	maximizes cross-modal correlation in whitened space. one of the unaligned baselines named in the proposal.
B2	Orthogonal Procrustes. classical, linear. finds the orthogonal rotation that best aligns the two modalities' PCA-projected spaces.
B3	Unaligned PCA. independent per-modality PCAs with L2-norm. no cross-modal objective. corresponds to the proposal's "unaligned baseline" sanity floor.
B4	Random-initialized untrained MLPs. R1 architecture with Xavier init, no training. strict baseline that isolates the contribution of contrastive training from the architecture alone.

metric definitions

metric	what it measures
R@K (recall at K)	for each query niche, rank candidates by cosine similarity. fraction of queries where the true match appears in the top K. proposal H1 retrieval metric.
MRR (mean reciprocal rank)	average of 1/rank across queries. rewards both high ranks and consistency. proposal H1 retrieval metric.
AUC	probability that a random matched pair has higher cosine similarity than a random mismatched pair. 0.5 = chance, 1.0 = perfect. scale-invariant. proposal H1 separability metric.

metric	what it measures
alignment gap	mean(matched cosines) - mean(mismatched cosines). positive = matched pairs systematically closer. proposal H1 separability metric.
CKA (Centered Kernel Alignment)	structural alignment between modalities: do the H&E and ST manifolds have similar pairwise- similarity structure overall? proposal H1 representation- structure metric.
ARI (Adjusted Rand Index)	cluster-vs-label agreement, corrected for chance. ARI=1 perfect, ARI=0 random. proposal H2 Part A metric (on the audit- correct mc_megacluster label, not the proposal-literal archetype label).
matched-pair cosine	within a tissue compartment, average cosine similarity between paired H&E and ST niches. proposal H2 Part B metric.
z_A	held-out cross-patient CCA correlation expressed in permutation-null standard- deviation units. proposal H3 metric, evaluated on cross- patient sub-split as the discriminating operationalization.
sig testable (/4)	per-run count of named-parent pathways passing BH-FDR<0.05 on the cross-patient sub-split. TGF-beta set_size=2 underpowered; excluded from FDR family. 4 testable pathways: Immune, ECM, Cell Cycle, PCD.
NMI(archetype, patient_id)	construct-validity diagnostic: how mechanically does the archetype label collapse to

metric	what it measures
	patient identity? 0 = independent, 1 = deterministic. observed 0.89 on 14 held-out patients (every niche in a patient inherits one archetype).

results by hypothesis

H1 · Cross-Modal Alignment

field	detail
Proposal hypothesis	Contrastive learning will improve cross-modal retrieval and representation alignment relative to unaligned baselines (CCA projection, late fusion concatenation, raw unaligned concatenation), as measured by retrieval metrics (R@K, MRR, median rank), alignment separability (alignment gap, AUC), and representation structure (CKA). Improvements expected within ~0.65-0.75 range, emphasis on relative improvement over baselines.
Evaluation question	Do contrastive runs (R1-R6) outperform unaligned and classical baselines (B1 CCA, B3 Unaligned PCA, B4 random-init) on retrieval, separability, and representation-structure metrics?
Construct-validity check	The H1 evaluation question is a clean operationalization of the hypothesis - retrieval and AUC are direct measurements of cross-modal alignment quality

field	detail
	and B4 random-init provides a strict chance-floor anchor. No re-framing needed.
Observed result	R4 AUC 0.859, CKA 0.631 (exceeds proposal range). R1 / R5 AUC 0.761 (top of range). R6 AUC 0.737 (in range). R2 0.725 (in range). R3 0.674 (slightly below range). Among classical baselines: B2 Procrustes 0.706, B1 CCA 0.544, B3 Unaligned PCA 0.441. B4 random-init at chance (0.496). Contrastive runs and classical baselines do not overlap on AUC.
Interpretation	Hypothesis supported. All 6 contrastive runs outperformed every baseline named in the proposal (B1, B3, B4). The CCA baseline (B1) underperformed contrastive runs by 0.13 to 0.32 AUC. B4 random-init at chance confirms contrastive training (not architecture) is doing the H1 work. R4 cross-attention exceeds the proposal range; late-fusion runs land inside or close to it. v3 build improved or held every contrastive run on AUC vs v1; classical baselines essentially unchanged.
Scientific conclusion	H1 holds. Contrastive alignment produces stronger cross-modal retrieval and representation alignment than unaligned baselines on TNBC-92. The proposal range was conservative for cross-attention; the alignment evidence is strongest there.
Primary metrics	R@K, MRR, AUC, CKA, alignment gap
Strong methods	R4 (best), R1, R5, R6, R2

field	detail
Baselines beaten	B1 (CCA), B3 (unaligned PCA), B4 (random-init); B2 Procrustes beaten by contrastive late-fusion runs and R4

H2 Part A · Structural Coherence (niche-level region clustering)

field	detail
Proposal hypothesis	The 9 spatial archetypes will cluster more coherently in the aligned latent space compared to individual modality spaces, measured by Adjusted Rand index and silhouette score.
Evaluation question (as proposed)	Do the 9 spatial archetypes cluster more coherently in the aligned space than in unimodal spaces? The proposal-literal evaluation question is not a good operationalization of the hypothesis. Wang's archetype labels were assigned at patient level: $NMI(\text{archetype}, \text{patient_id}) = 0.89$ on the 14 held-out patients; every niche in a patient inherits one archetype.
Construct-validity check	Archetype ARI mechanically resolves to a 14-class patient classifier at niche resolution. Classical baselines that amplify patient identity (B1 CCA: patient $z = 122.13$, $3.3\times$ amplification over raw _{he}) score highest on the proposal-literal metric without encoding more biology. Corrected evaluation question: do niches cluster by transcriptomic tissue state, using

field	detail
Observed result	Wang's per-spot 14-class <code>mc_megacluster</code> (NMF) label as the audit-correct niche-resolution biology target? KMeans(k=14) on <code>z_he</code> against <code>mc_megacluster</code> (corrected label): R5 ARI 0.246, R6 0.244, R1 0.234 (3-way tie within KMeans seed noise, gaps <0.012). R4 0.101 (lowest on KMeans). Classical baselines lose 11+ ARI points on the resolution shift archetype -> <code>mc</code> (B2/B3 -0.112, B1 -0.048), quantifying patient-leakage drop. Raw ST alone: 0.046. Aligned space provides 5× lift over raw ST. R4 rescue confirmed on v3: patient-stratified linear probe (LogReg 14-class, 11/3 patient split) gives R4 accuracy 0.240, at parity with R1 (0.282), R6 (0.276), <code>raw_he</code> (0.270) - the KMeans-low result is manifold-geometry bias (cross-attention compresses to a tight non-globular manifold; KMeans assumes globular clusters; LogReg only needs linear separability), not biology absence. kNN purity R4 0.628 vs ~0.80 reflects R4's patient suppression, not biology loss.
	On the construct-validity-corrected label, the hypothesis is supported. Aligned latent space clusters niches by transcriptomic tissue state more coherently than raw ST alone (5× ARI lift). Late-fusion contrastive runs (R1, R5, R6) win the niche-level region clustering by KMeans; R4's tight cross-attention manifold is poorly served by KMeans's
Interpretation	

field	detail
	globular-cluster assumption but encodes mc biology at linear-probe parity (R4 0.240 vs R1 0.282, raw_he 0.270) - confirmed on v3, matching the v1 rescue. R6 wins parsimony - it drops gpath2vec from the ST input entirely and Novae's spatial-neighborhood graph alone clusters mc states. H2 Part A holds on the audit-correct label. The proposal-literal archetype label is reported as diagnostic only and visualized in notebook §5 alongside patient_id panels to show the construct-validity collapse. Aligned space organizes niches by transcriptomic tissue state at a resolution that raw ST embedding cannot reach.
Scientific conclusion	
Primary metrics	mc_megacluster ARI (z_he), NMI(archetype, patient_id) for construct-validity, archetype -> mc delta for patient-leakage diagnosis
Strong methods	R5, R6, R1 (3-way tie within seed noise)
Baselines	B1, B2, B3 lose 11+ ARI points archetype -> mc, quantifying the patient-leakage component

H2 Part B · Compartment / FTU Coherence (same FTU, two ground truths)

H2 Part B tests the FTU-coherence aspect of the proposal - does the aligned space preserve compartment-level / functional-tissue-unit organization? The proposal names two relevant ground truths: the **17 expert-annotated morphological categories** (categorical pathologist labels including “Lymphoid nodule” = TLS) and the **TLS gene**

signatures scored spatially (continuous per-niche TLS-signature score). Both target the same FTU (TLS), at different resolutions of the same biology. We test both and report them as **B.1** (categorical) and **B.2** (continuous). The two tests give different method rankings - see the framing note after B.2 for what that tells us about what alignment preserves.

H2 Part B.1 · categorical compartment cosine contrasts (17-class pathologist annotations)

field	detail
Proposal hypothesis	Alignment quality varies across TME compartments: morphologically distinct compartments (tumor, TLS, necrosis) will show higher matched-pair cosine similarity than morphologically ambiguous compartments (high-TIL vs low-TIL stroma).
Evaluation question	Do tumor / TLS / necrosis compartments produce higher matched-pair cosine than high-TIL / low-TIL stroma, measured by Welch z-test on the 6 directional contrasts?
Construct-validity check	The compartment-level test is conceptually clean: per-spot pathologist annotations (17 categories, 12 populated in test set) vary at niche resolution unlike the patient-level archetype label. Caveat: the Tumor (n≈4384) and Necrosis (n≈1178) annotation classes pool heterogeneous subtypes and dilute within-compartment cosine; small well-curated TIL annotations (n=105 to 354) have higher within-compartment homogeneity. The directional prediction is sensitive to annotation granularity, not just biology.
Observed result (v3)	Welch z per run (one-sided, distinct > ambiguous as proposal predicts) on

field	detail
	runs/tnbc-92_v3/eval/H2/ compartments_welch_contrasts.parquet. TLS vs hi-TIL stroma: pass in R6 (+5.6), R2 (+5.4), B2 (+3.7), R5 (+2.4), R1 (+2.2). TLS vs lo-TIL stroma: pass in R6 (+5.9), R2 (+5.4), R1 (+4.1), R5 (+3.2), B2 (+2.8). Tumor and Necrosis contrasts invert decisively in 4-7 runs each (z down to -15). R3 uniquely passes Necrosis contrasts (z = +5.8, +9.1). R4 fails every contrast (0/6 pass, 3 inverted) - its tight cross-attention manifold collapses compartment distinctions. v3 coverage: 13/38 held-out test subarrays carry compartment annotations (~35% of test niches). The TLS direction held cleanly. TLS is the one well-curated FTU annotation in the dataset and the contrast where the proposal's directional prediction matched. Tumor and Necrosis annotations are too heterogeneous to differentiate from TIL stroma on matched-pair cosine; this is an annotation-vocabulary issue. Late-fusion contrastive runs (R1, R2, R5, R6) preserve compartment-cosine separation on TLS - exactly the matched-pair structure their loss optimizes. R4 sacrifices it for the cross-attention manifold compression. R3 (Barlow Twins) uniquely picks up Necrosis. H2 Part B.1 holds for TLS. The categorical compartment-cosine test supports the FTU-coherence claim on the one well-curated FTU annotation. The Tumor and Necrosis arms are not testable as proposed without annotation-vocabulary refinement.
Interpretation	
Scientific conclusion	
Primary metrics	

field	detail
	Matched-pair cosine by compartment, Welch z (one-sided) on 6 named contrasts
Strong methods	R6, R2, R1, R5, B2 (TLS contrasts); R3 (Necrosis contrasts, unique)
Baselines	B1 (1/6 pass; mostly non-significant); B3 (0/6 pass - sanity floor, no compartment differentiation)

H2 Part B.2 · continuous TLS gene-signature CCA

field	detail
Proposal hypothesis	TLS gene signatures scored spatially are one of the four ground-truth sources listed in the proposal for evaluating whether aligned representations capture shared biological signal across modalities.
Evaluation question	Does the H&E view of each run's aligned space (z_{he}) linearly predict the continuous per-niche TLS gene-signature score (from <code>tls_scores.tsv</code>) on held-out patients?
Construct-validity check	This is a cross-modal prediction question: H&E never sees the gene-signature score during training (the signature is computed from ST counts). z_{he} is the citable non-circular view. z_{st} and z_{mean} are partially circular for the 8 runs that include <code>gpath2vec</code> on the ST input - they're reported in the parquet but not as headline. The continuous test complements the categorical B.1: same FTU (TLS), different operationalization, answers a different question - <i>"does z_{he} linearly decode the TLS</i>

field	detail
	<i>biology?” vs B.1’s “does the aligned space preserve matched-pair structure within the TLS compartment?”</i> B4_v3 (random-init) z_TLS 25.46 (strongest). B1_v3 (CCA) 16.61. R4_v3 (cross-attn) 11.39. B3_v3 / B2_v3 (rotation-equivalent classical) 11.29. R5_v3 marginal at 2.37 (passes FDR but borderline). R6_v3 1.54, R3_v3 0.54, R1_v3 -0.17, R2_v3 -0.78 - all non-significant. 6 of 10 runs sig on z_he. TLS gene signature is directly H&E-morphology-decodable - organized lymphoid aggregates have a distinctive Virchow2-feature signature, and a random Xavier projection of those features + L2-norm (B4) preserves the cross-patient predictive signal end-to-end. Classical linear methods (B1, B2, B3) and R4 cross-attention also preserve it. Late-fusion contrastive runs (R1, R2, R3, R6) actively degrade the linear-decodability of TLS on z_he - the contrastive objective distorts the H&E features in ways that help cross-modal matching (the H1 / H3 wins) but hurt pure-morphology decoding tasks like this. R5’s per-dim temperature lets it retain a marginal signal (z=2.4). H2 Part B.2 supported for the H&E-feature-preserving methods (random init, classical baselines, R4). Late-fusion contrastive runs fail this signal -
Observed result (v3, z_he, cross-patient held-out 3-patient sub-split, 500 perms, BH-FDR<0.05)	
Interpretation	
Scientific conclusion	

field	detail
	this is a methodologically informative <i>negative</i> finding: contrastive alignment training has a cost on signals already strongly encoded by a single modality's foundation features.
Primary metrics	univariate CCA on z_{he} , cross-patient held-out 3-patient sub-split, 500 perms, BH-FDR<0.05; $z_{TLS} = (\text{held-out test } r - \text{null mean}) / \text{null std}$
Strong methods	B4 (best), B1, R4, B2 / B3, R5 (marginal)
Baselines	R1, R2, R3, R6 below significance - the <i>contrastive late-fusion family fails this signal</i> despite passing B.1

What B.1 vs B.2 tells us (framing)

The two tests give different rankings because they answer different questions about the same FTU.

	B.1 (categorical compartment cosine)	B.2 (continuous TLS gene-signature CCA on z_{he})
what it tests	are matched (H&E, ST) pairs closer within the TLS compartment than to other compartments?	does z_{he} linearly decode the continuous TLS biology score in held-out patients?
strong methods	late-fusion contrastive (R1, R2, R5, R6) + B2 Procrustes	morphology-preserving (B4 random-init, B1 CCA, R4 cross-attn, B2/B3)
failing methods	R4 (tight cross-attn manifold), B1 / B3 (no compartment differentiation)	late-fusion contrastive (R1, R2, R3, R6)

The two TLS tests answer different questions about what alignment preserves. B.1 rewards methods whose loss optimizes within-compartment matched-pair cosine - that's the contrastive late-fusion family by construction. B.2 rewards methods that preserve linear decodability of a signal already encoded in raw H&E features - that's any method that doesn't distort Virchow2's morphology axes (random init, CCA, Procrustes). Contrastive training is **not free**: it pays a cost on signals already morphology-decodable (like TLS in raw Virchow2) in exchange for the cross-modal coupling that drives H1 and H3 cross-patient pathway transfer. The TLS FTU is supported either way - but the *which method* answer is task-conditional, which is the systems-oriented hypothesis instantiated on a specific signal.

H2 Part C · Biology vs Patient Signal (added analysis)

field	detail
Hypothesis (extension of H2)	Aligned space encodes biology more strongly than patient identity, measured by permutation-null z-scores on Bareche TIME / MC_global / MC_tumor signatures and ratio of bio z to patient z. Added to address whether H1 retrieval gains reflect biology or patient-identity leakage.
Evaluation question	Does the aligned space encode biology more strongly than patient identity? bio z / patient z ratio is the headline metric.
Construct-validity check	The bio-vs-patient ratio directly measures the failure mode that flagged the archetype label - if a run encodes patient identity more strongly than biology, it can win patient-label-collapsed metrics without encoding real biology. The ratio operationalizes "is this run useful for cohort-

field	detail
	level interpretation, or is it patient-leakage in disguise?”
	bio/patient ratio = TIME-signature bio z / patient z (higher = biology dominates). R6 0.547 (highest), R5 0.497 , R1 0.461, R4 0.376, R2 0.372, R3 0.250, B1 0.071 (anti-helpful). R4 has the lowest patient z (21.5, strongest patient suppression); B1 amplifies patient identity ~3× (patient z 113.0 vs raw_he 36.8). raw_he floor = 0.137, raw_st floor = 0.088.
Observed result (v3, z_he, 10k perms)	All six contrastive runs sit above the raw_he floor; B1 (CCA) alone falls below it. Reproduces v1 (R6 0.548, R1 0.517, B1 0.064, raw_he 0.129). v3 notebook §6 visually confirms - B1 panels show tight per-patient clusters; R4 panels mix patients best.
Interpretation	Aligned-space embeddings preserve biological organization while suppressing patient-specific signal, except for B1 CCA. R4’s strong patient suppression on z_he is what enables its H3 cross-patient pathway transfer (see below). B1 amplifies patient identity more than biology and should not be used for any analysis where cross-patient validity matters.
Scientific conclusion	H2 Part C supported for contrastive runs; B1 fails. All six contrastive runs encode TIME/MC biology above the raw-modality floor (ratios 0.25-0.55); B1 (CCA) amplifies patient identity ~3× and falls below the floor (0.071) - anti-helpful for

field	detail
	cohort-level interpretation. v3 reproduces the v1 verdict numerically.
Primary metrics	Patient-ID decoding, bio z / patient z ratio on Bareche TIME / MC_global / MC_tumor signatures
Strong methods	R4 (patient suppression), R6 (bio/patient ratio), R1 (bio/patient ratio)
Baselines	B1 (anti-helpful in v1)

H3 · Pathway Interpretability

field	detail
Proposal hypothesis	5 Reactome pathways correlate with shared-latent directions via CCA, indicating preservation of interpretable biological signal.
Evaluation question	Do the 5 named Reactome pathway embeddings correlate with directions in each run's shared latent space, AND does the correlation transfer across held-out patients?
Construct-validity check	Two operationalizations of the hypothesis: (1) within-cohort fit-on-all CCA, (2) cross-patient sub-split CCA. Operationalization (1) is the proposal-literal test but does not discriminate between alignment strategies (universally positive). Operationalization (2) - cross-patient sub-split - is the discriminating test that separates pathway encodings that transfer from those that overfit to patient-specific gradients. We report (2) as the headline. Circularity caveat: v3 gpath2vec is on the ST input, so z_st / z_mean views are

field	detail
	circular with the gpath2vec-derived signal. Only z_he is non-circular and reported as headline; z_st / z_mean are in the parquet for completeness. Option A cross-patient held-out (3-patient sub-split; 500 perms; BH-FDR over testable pathways), z_he view. R4 4/4 testable significant, z_A range Immune 25.4, ECM 16.0, Cell Cycle 16.5, PCD 11.2. Classical baselines 4/4 with lower z_A: B1 range 7.5-15.2, B2/B3 6.7-17.1. Late-fusion contrastive 0-2/4: R5 2/4 (ECM 3.0, CC 2.1, PCD 3.0; Immune actively suppressed at z_A -1.9), R1 1/4, R3 1/4, R2 0/4, R6 0/4. TGF-beta R-HSA-170834 set_size=2, underpowered, excluded from BH-FDR family. v3 R4 z_A is stronger than v1 R4 on the AUCell arm (v1 R4 z_A 3.8-5.7; v3 gpath2vec-arm R4 z_A 11.2-25.4). Hypothesis supported for R4 and classical baselines on the discriminating cross-patient test. Late-fusion contrastive runs encode pathway signal that does not transfer - they fit pathway-correlated directions within-cohort that are patient-specific in structure. R4's pathway-direction off-diag mean ~0.75 on z_he means the 5 named pathway directions collapse toward one biology axis; the aliasing IS the source of cross-patient generalization (one axis is harder to overfit to patient identity than five distinct directions). Classical baselines
Observed result	
Interpretation	

field	detail
	preserve per-pathway distinguishability with lower magnitude. H3 holds for R4 and classical baselines on the proposal-literal pathway CCA test, evaluated cross-patient. The aligned shared latent preserves interpretable pathway signal that transfers to held-out patients in the cross-attention and classical families. Late-fusion contrastive runs need additional treatment (different objective or architecture) to encode cross-patient transferable pathway signal; H3 does not hold for that family on this cohort. Routing rule (unchanged from v1): R4 for cross-patient magnitude on any one pathway; B2 for per-pathway distinguishability across multiple pathways.
Scientific conclusion	
Primary metrics	Per-pathway univariate CCA z_A on z_{he} , permutation null (500 perms), BH-FDR over 4 testable pathways
Strong methods	R4 (overall winner 4/4); B1/B2/B3 (classical fallback 4/4 with lower z_A)
Notes	DAG-resolution H3 decomposition (395 sub-pathway nodes; v1 had R4 9.9% sig across 9480 cells with biologically distinct top hits including BTLA $z=6.92$, Integrin $z=6.21$, Collagen $z=5.22-5.28$) is deferred for v3. v1 numbers preserved at ../v1/tnbc92_results_summary.md.

Systems-Oriented Evaluation

field	detail
Proposal framing	Multimodal alignment as an orchestration and evaluation problem rather than a single-model optimization problem. The proposal predicted that different alignment strategies would optimize different biological objectives.
Evaluation question	Does any one alignment strategy dominate every hypothesis family simultaneously on the v3 retrain?
Construct-validity check	The hypothesis is about the absence of a global winner. The risk is that the metric grid is degenerate (all hypotheses measure the same underlying property). The H1/H2/H3 grid passes this check: H1 measures cross-modal coupling, H2 Part A measures niche-level transcriptomic clustering, H2 Part B measures compartment-level FTU coherence, H3 measures cross-patient pathway transfer. These are not co-monotonic - a run can win one and lose another, which is exactly what is observed.
Observed result	No single method dominates every hypothesis family on v3. R4 wins H1 cross-modal retrieval and H3 cross-patient pathway transfer. R5 / R6 / R1 tie at the top on H2 niche-level mc_megacluster clustering. Classical baselines fail H1 retrieval (B1 0.544 vs R4 0.859) but stay competitive on H3 cross-

field	detail
	patient hit-rate with lower z_A magnitude. B1 amplifies patient identity (v1 result, v3 visual confirmation in notebook §6). Hypothesis supported as proposed. Niche-level cluster geometry, cross-modal retrieval, and cross-patient pathway transfer are not co-monotonic objectives. R4's tight aliased manifold trades cluster geometry (H2 Part A) for retrieval + pathway transfer (H1 + H3). R5/R6/R1's wider manifold does the reverse. The trade-off is real and reproducible (v1 -> v3) and is the framework's value-add for downstream routing.
Interpretation	
Scientific conclusion	The systems-oriented framing holds: omicstra's role is to expose the trade-offs so downstream analyses route the right question to the right method. v3 retrain reproduces the v1 directional verdict end-to-end.
Primary metrics	Integrated evaluation across H1, H2 (corrected), H3 (cross-patient)
Strong methods	R4 (retrieval + cross-patient pathway transfer); R5 / R6 / R1 (niche-level region clustering)
Baselines	all baselines lose at least one hypothesis family decisively; B1 fails on H2 construct-validity diagnostic and H3 amplification check

task-conditional routing rule (v3)

the H1/H2/H3 verdicts collapse to this routing rule for downstream use:

if the question is	route to	evidence (v3)
cross-modal retrieval (find matched ST niche from H&E)	R4	H1 AUC 0.859, CKA 0.631, alignment gap +0.233
group niches that share similar tumor- microenvironment state (immune-hot, stromal, necrotic, TLS, etc.) to find biologically meaningful regions in a tissue section	R5, R6, or R1 (3-way tie within seed noise)	H2 Part A KMeans on mc_megacluster (audit-correct label, 14-class NMF): R5 ARI 0.246, R6 0.244, R1 0.234
amplify biology, suppress patient identity for cohort- level interpretation	R6 or R5 (highest bio/ patient ratio); R4 (lowest patient z, leakage minimization)	v3: R6 ratio 0.547, R5 0.497, R1 0.461; R4 patient z 21.5 (lowest in grid)
find pathway-similar niches across new patients	R4	H3 cross-patient gpath2vec arm: 4/4 testable pathways, z_A range 11.2-25.4
interpret a single pathway's transferable signal cross-patient	R4 (with B1/B2 classical fallback)	H3 R4 z_A 25.4 on Immune; B2 z_A 17.1 on Immune; R4 wins magnitude, classical match hit-rate
interpret per- pathway distinguishability across multiple pathways	B2 (Procrustes)	classical preserves per-pathway directional separation; R4 aliases the 5 named pathway directions onto ~one axis (off-diag ~0.75 on z_he)
avoid for cohort-level biology interpretation	B1 (CCA)	construct-validity diagnostic - v3 B1 patient z 113.0, ~3×

if the question is	route to	evidence (v3)
		amplification; bio/ patient ratio 0.071 below raw-modality floor 0.137 (anti- helpful)

no single method dominates every objective; the framework's value is in exposing the trade-offs so that downstream analyses route the right question to the right method. This is what the omicstra alignment-agent encodes as task-conditional dispatch. When this evaluation runs on additional cohorts (HEST, future TNBC studies, other tumor types) and each run's final matrix is persisted back to the alignment-agent's rules store, a routing pattern accumulates: which question types map reliably to which method families, where the trade-offs hold, and where they shift with cohort characteristics. TNBC-92 v3 is the second row.

consolidated proposal-alignment matrix (v3)

hypothesis	proposal statement	evaluation question	construct validity	observed result (v3)
H1 Cross-Modal Alignment	Contrastive learning improves cross-modal retrieval and representation alignment over unaligned baselines (CCA, late fusion, raw). Expected range 0.65-0.75 on cross-modal similarity.	Do contrastive runs outperform B1 CCA, B3 unaligned PCA, B4 random-init on R@K / MRR / AUC / alignment gap / CKA?	Clean operationalization - retrieval and AUC directly measure alignment quality; B4 chance is a strict floor.	R4 AUC 0.859, CKA 0.631 (above range). R1/R5 0.761, 0.737, R2 0.72 (in range). R3 0.674 (slightly below). B2 0.706, B1 0.54, B3 0.441, B4 0.491. No overlap between contrastive and classical.

hypothesis	proposal statement	evaluation question	construct validity	observed result (v3)
H2 Part A Structural Coherence	9 spatial archetypes cluster more coherently in aligned space than in unimodal spaces (ARI, silhouette).	(proposal-literal) Do archetypes cluster more coherently? (corrected) Do niches cluster by transcriptomic state on Wang's per-spot mc_megacluster label?	Proposal-literal question fails construct validity - archetype is patient-level pseudobulk (NMI 0.89 vs patient_id); ARI mechanically resolves to patient classification. B1 CCA wins by amplifying patient identity (z=122.13). Correction: use mc_megacluster (audit-correct, niche-resolution NMF label) as the biological target.	KMeans on mc_megacluster (corrected): R5_v3 0.246, R6_v3 0.244, R1_v3 0.234 (way tie, seed noise). R4_v3 0.101. Classic baselines lose 11+ ARI archetype -> (B2/B3 -0.112, B1 -0.048), quantifying patient-leakage. Raw ST alone 0.046. Aligned space 5x lift over raw ST. KMeans-low manifold-geometry bias v3 linear projection gives R4 0.24 at parity with R1 0.282 / raw_he 0.270 confirming biology is present (not absent).
H2 Part B.1 Compartment Cosine (categorical FTU annotations)	Tumor / TLS / necrosis show higher matched-pair cosine than high-TIL / low-TIL stroma.	Do TLS / Tumor / Necrosis compartments produce higher matched-pair cosine than high-TIL / low-	Per-spot pathologist labels vary at niche resolution (unlike archetype). Tumor / Necrosis annotation classes are heterogeneous and dilute	v3 Welch z: T contrasts pass in R6/R2/R1/B2 (z = +2.2 to +5.9 on both ambiguous-stroma comparisons Tumor /

hypothesis	proposal statement	evaluation question	construct validity	observed result (v3)
		TIL stroma on Welch z-tests?	matched-pair cosine - granularity caveat.	Necrosis inv in 4-7 runs. R uniquely pas Necrosis (z = +5.8, +9.1). R fails every contrast (0/6), v3 coverage: 13/38 subarrays.
H2 Part B.2 TLS Gene Signature (continuous - the 4th proposal ground truth)	TLS gene signatures scored spatially are one of the four proposal-named ground truths for evaluating preserved biological signal.	Does z _{he} linearly predict the continuous per-niche TLS-signature score on held-out patients via univariate CCA + permutation null?	z _{he} is the non-circular view (H&E never sees the gene-signature score, which comes from ST counts). The continuous test complements the categorical B.1 - same FTU, different operationalization.	v3 z _{he} cross-patient sub-split, 500 perms, BH-FDR<0.05: B4 random-init z_TLS = 25.5 (strongest), B CCA 16.6, R4 cross-attn 11 B2/B3 11.3, R 2.4 (marginal R6/R3/R1/R2 (z_TLS < 1.6, ns). 6 of 10 ru sig.

hypothesis	proposal statement	evaluation question	construct validity	observed result (v3)
H2 Part C Biology vs Patient (extension)	Aligned space encodes biology more strongly than patient identity.	bio z / patient z ratio on Bareche TIME / MC_global / MC_tumor signatures; patient-ID decoding.	Directly measures the failure mode flagged by H2 Part A audit - is the run patient-leakage in disguise?	v3 (z_he, 10k perms): R6 raw 0.547, R5 0.49, R1 0.461, R4 0.376, R2 0.37, R3 0.250 - all above raw_h floor 0.137. B 0.071 (below floor; patient 113.0, ~3× amplification R4 lowest patient z (21. Reproduces v (R6 0.548, B1 0.064).
H3 Pathway Interpretability	5 Reactome pathways correlate with shared-latent directions via CCA, indicating preservation of interpretable biological signal.	(proposal-literal) within-cohort CCA. (discriminating) cross-patient sub-split CCA z_A.	Fit-on-all is universally supported, does not discriminate. Cross-patient sub-split separates transferable from patient-overfit pathway encodings - report as headline. z_he is the non-circular view (v3 gpath2vec on ST input makes z_st / z_mean circular).	Cross-patient z_he, BH-FDR<0.05: R4_v3 4/4 (z_ 11.2-25.4). Classical B1/I B3 4/4 (z_A 3-17). R5_v3 2 R1/R3 1/4. R2 0/4. TGF-beta set_size=2 underpowered v3 R4 z_A stronger than v1 R4 on AUOC arm.

hypothesis	proposal statement	evaluation question	construct validity	observed result (v3)
Systems-Oriented	Multimodal alignment as orchestration and evaluation, not single-model optimization. Different strategies should optimize different objectives.	Does any one alignment strategy dominate every hypothesis family on v3?	H1/H2/H3 grid is non-degenerate - measures different underlying properties. A run can win one and lose another.	No. R4 wins L + H3. R5/R6/E tie H2 Part A Classical fail retrieval, competitive c H3 hit-rate. E amplifies patient ident

Pointers. v3 alignment runs: runs/tnbc-92_v3/{R1,R2,R3,R4,R5,R6,B1,B2,B3,B4}_v3/. **v3 H1:** runs/tnbc-92_v3/eval/H1/summary.json. **v3 H2:** runs/tnbc-92_v3/eval/H2/{summary.json, mc_coherence.parquet, compartment_cosine.parquet}. **v3 H3 gpath2vec arm:** runs/tnbc-92_v3/eval/H3/pathway_cca_gpath2vec_v3/{per_pathway_cca.parquet, perm_nulls.parquet, provenance.json}. **v3 H3 AUCell (orthogonal robustness check):** runs/tnbc-92_v3/eval/H3/pathway_cca/. **v3 summary notebook + HTML:** notebooks/final/{05_summary_umaps_v3.ipynb, 05_summary_umaps_v3.html}. **v1 baseline (predecessor):** ../v1/tnbc92_results_summary.md. **construct-validity audit:** projects/tnbc-92/evaluation_question_audit.md. **MCP / LangGraph agent design:** docs/mcp_agent_design.md.

draft-for-voice-pass. prose is Claude-drafted scaffolding (per memory feedback_writing_voice); every number verified against runs/tnbc-92_v3/eval/*; voice pass needed before publishing.